BFHL HealthPay

Recon & Settlement Automation

Design Thinking-Led Product Case Study

Final Confirmed Version

> *"HealthPay transformed reconciliation from manual cross-system checking into an automated, controlled financial operations layer for payment matching, benefit validation, settlement proofing, payout triggering, exception management, and audit-ready reconciliation."*

Document Attribute	Final Detail
Project	HealthPay Recon & Settlement Automation
Year	2023
Business unit	HealthPay
Timeline	Immediately after the Service Guarantee project
Role	Lead Product Manager for the full project
Scope	Pay-in and pay-out automation for doctor and lab payouts
Final payout rail	RazorpayX
Primary success metrics	Payout TAT and settlement accuracy

Source artifacts reviewed: Recon Automation PayIn and PayOut.docx; Recon and Settlement SOP.xlsx; US Descriptions.docx

Contents

1. Executive Summary

HealthPay Recon & Settlement Automation was a 2023 HealthPay initiative executed immediately after the Service Guarantee project. The project focused on automating complex pay-in and pay-out reconciliation across customer payments, benefits, payment gateway settlements, finance receipts, doctor payouts, and lab payouts.

As Lead Product Manager, Samadhan owned the full project, including PRD, user stories, dashboard design, reconciliation logic, manual review workflows, audit controls, payout-triggering requirements, and daily operations review until stabilization.

The solution went live across all key capabilities: dashboard, raw data ingestion, API ingestion from all sources, manual upload fallback, matching engine, manual review, audit log, and payout triggering through RazorpayX. The process became completely automated, eliminated manual reconciliation effort, enabled T+1 payouts, reduced payout delays and provider escalations, and improved settlement accuracy, auditability, compliance, and financial controllership.

> *"The core product shift was from manual reconciliation and payout tracking to system-led financial matching, exception surfacing, controlled correction, maker-checker approval, and audit-ready settlement operations."*

Area	Final Summary
Project type	Financial operations, reconciliation, settlement, and payout automation
Primary users	Finance, operations, settlement teams, product/business users, and provider payout stakeholders
Primary metrics	Payout TAT and settlement accuracy
Core outcome	Manual reconciliation effort eliminated; payouts happened on T+1
Governance outcome	Role-based edits, full audit logs, and maker-checker controls

2. Confirmed Project Context

Attribute	Confirmed Detail
Organization / Unit	HealthPay
Project year	2023
Chronology	Immediately after BFHL Service Guarantee
Product scope	Pay-in and pay-out reconciliation automation
Payout coverage	Doctor payouts and lab payouts
Final payout rail	RazorpayX
Lead PM ownership	Full project ownership
Artifacts owned	PRD, user stories, dashboard design, solutioning
Operations involvement	Daily operations review until stabilization
API ingestion	Went live for all sources
Manual uploads	Retained as fallback
Go-live capabilities	Dashboard, raw data ingestion, API ingestion, manual upload, matching engine, manual review, audit log, payout triggering
Governance	Role-based edit permissions, full manual correction audit logs, maker-checker approval for edited raw data/manual payout

3. Strategic Framing

A weak framing would be: Built a reconciliation dashboard.

That undersells the project. This was a cross-system financial operations automation initiative where payment, benefit, settlement, and payout data needed to reconcile across multiple systems before doctor and lab payouts could be triggered safely.

> *"Led HealthPay Recon & Settlement Automation as Lead PM, delivering API-led reconciliation, matching engine, manual review, audit logs, maker-checker controls, dashboarding, and RazorpayX payout triggering across pay-in/pay-out flows for doctor and lab payouts."*

4. Systems Landscape

System	Role in HealthPay Recon / Settlement
PayU	Payment gateway used for off-us QR transactions and pay-in validation.
RazorPay	Payment gateway used for on-us transactions, especially where net customer payment is collected after benefits.
PayRx	Central payment/reconciliation system storing transaction and payout context.
HRx	Source of benefit/payment context and appointment-related commercial data.
DRx	Doctor/lab booking and benefit breakup context; passes voucher, Healthcoin, claim, appointment, and provider details.
Labs Backend	Lab-side appointment or transaction records used for reconciliation and lab payout validation.
Voucherify	Voucher redemption and voucher amount source.
Healthcoin	Healthcoin redemption and benefit amount source.
Claims	Claim benefit component used as part of transaction value and payout calculation.
Finance / Bank Statement	Settlement credit validation and financial closure source.
RazorpayX	Final payout rail for doctor and lab/provider settlement.

5. Core Design Thinking Insight

The core insight was that HealthPay was not handling simple single-ledger payments. A single HealthPay transaction could include customer net payment, voucher amount, Healthcoin amount, claim benefit, payment gateway charges, taxes, bank settlement credit, and final provider payout.

> *"From payment capture to financial truth reconciliation."*

The right product questions were not limited to whether payment happened. The system had to answer:

- Did every source system record the same transaction?
- Did the payment gateway record match PayRx?
- Did the gross transaction value equal net paid plus voucher, Healthcoin, and claim benefit?
- Did Voucherify, Healthcoin, Claims, DRx, HRx, and PayRx agree on the benefit breakup?
- Did PG settlement minus fees and taxes match the bank credit?
- Did provider payout reflect the complete payable value?
- Can exceptions be identified, reviewed, corrected, approved, and audited?

6. Business and Operational Problem

HealthPay finance and operations teams needed an automated way to reconcile pay-in, benefit utilization, settlement, and payout records across multiple systems because transaction data, benefit components, gateway settlements, bank credits, and provider payouts were distributed across different ledgers and operational systems.

Problem Type	How It Appeared in HealthPay
Data Fragmentation	Payment, benefit, booking, settlement, and payout data existed across PayU, RazorPay, PayRx, HRx, DRx, Labs, Voucherify, Healthcoin, Claims, Finance, and RazorpayX.
Manual Recon Effort	Count and amount validations required system-to-system matching and exception review.
Settlement Risk	PG settlement, service fee, tax, net credit, and bank statement had to agree.
Benefit Mismatch Risk	Voucher, Healthcoin, and claim benefit amounts had to reconcile against PayRx/DRx/HRx records.
Payout Accuracy Risk	Doctor and lab payouts depended on net customer payment, benefits, and commercial/PG charge components.
Audit Risk	Raw data edits, manual corrections, and payout decisions needed traceability and approval.

Problem Type	How It Appeared in HealthPay
Provider Experience Risk	Payout delays and unresolved mismatches could create provider escalations.

7. Design Thinking-Led Discovery

The discovery lens was operational and financial rather than UI-first. The team had to understand how finance, operations, settlement teams, product/business users, and payout stakeholders worked with fragmented financial data.

Observed / Inferred Operating Reality	Product Interpretation
Transactions originated across PayU and RazorPay	Recon could not depend on one payment gateway ledger.
PayRx stored transaction and payout context	PayRx needed to become the operational reconciliation layer.
Benefits came through vouchers, Healthcoins, and claims	The matching engine needed multi-ledger value composition.
Doctor and lab payouts were dependent on accurate reconciliation	Payout triggering had to be controlled and auditable.
Manual review was unavoidable for exceptions	The product needed exception queues, not blind automation.
Raw data could require correction	Role-based edit permissions and full audit logging were essential.
Operations review continued until stabilization	The solution needed to work in real operating rhythm, not only in test scenarios.

8. Problem Statements

8.1 Core Problem Statement

> *"HealthPay finance and operations teams needed an automated and audit-ready way to reconcile pay-in, benefit utilization, settlement, and payout records because payment, benefit, settlement, bank credit, and provider payout data existed across multiple systems, resulting in manual reconciliation effort, payout delays, settlement risk, provider escalations, and weak controllership."*

8.2 Finance / Settlement Team Problem Statement

Finance teams needed reliable settlement validation because PG charges, service tax, net settlement amount, bank credit, and system transaction value had to match before financial closure.

8.3 Operations Team Problem Statement

Operations teams needed automated exception identification and manual review queues because count mismatches, amount mismatches, failed payouts, invalid bank/UPI details, and beneficiary bank downtime required structured intervention.

8.4 Provider / Doctor / Lab Problem Statement

Doctors and labs needed timely and accurate payouts because payout delays and unresolved mismatches could trigger provider escalations and trust issues.

9. Pay-In Reconciliation Scope

9.1 Off-Us Transaction: PayU to PayRx

22. Customer initiates an off-us transaction by scanning a QR.
23. Transaction goes through PayU.
24. PayU confirms the transaction to PayRx.
25. PayRx stores the transaction.
26. System matches transaction count and amount using PayRx Transaction ID / PG Order ID.
27. Any mismatch is flagged for review.

9.2 On-Us Transaction: RazorPay to PayRx

28. Customer initiates an on-us transaction by scanning a QR.
29. Customer may apply voucher, HealthCoin, and/or claim benefit.
30. HRx sends voucher/Healthcoin/claim context to DRx.
31. DRx sends benefit details to PayRx.
32. Net customer-paid amount is collected through RazorPay.
33. RazorPay confirms transaction to PayRx.
34. PayRx stores RazorPay transaction details.
35. System validates the full transaction value across net paid and benefit components.

> *"Gross transaction amount = Net paid by customer + Voucher + Healthcoin + Claim benefit."*

Payment Case	Reconciliation Rule
Full PG payment	Match transaction count and amount using PG Order ID / PayRx Transaction ID.
Partial PG payment	RazorPay amount should equal PayRx order amount minus Voucherify, HealthCoin, and Claim components.
Benefit-only payment	DRx benefit details should match Voucherify, Healthcoin, and Claims using unique parameters.

10. Benefit and Multi-Ledger Reconciliation

The project had to reconcile both cash and non-cash value components. This made the problem more complex than standard payment gateway reconciliation.

Benefit Component	Reconciliation Need
Voucherify	Voucher code, status, amount, redeem date, customer phone number, SRC_ID, and transaction count should match DRx/HRx/PayRx context.
Healthcoin	Healthcoin redemption amount and count should match DRx/HRx/PayRx context.
Claims	Claim benefit amount should be included in benefit breakup and gross transaction value.
DRx / HRx	Appointment value, voucher, Healthcoin, claim, net payable, PayRx ID, and SRC_ID should reconcile against PayRx and source systems.
Partial payment with benefits	PayRx order amount should equal RazorPay payment amount plus Voucherify amount plus Healthcoin amount plus claim amount.

The matching engine therefore needed to understand value composition, not only a single amount field.

11. Settlement and Payout Logic

11.1 PG Settlement vs Finance

The settlement validation required payment gateway settlement reports to reconcile against finance/bank statement credits.

Component	Validation Intent
PG gross amount	Total payment gateway captured amount.
Service fee	Gateway fee deducted from gross amount.
Service tax	Tax deducted/charged on gateway fee.
Net settlement amount	Expected bank credit after fees and taxes.
Bank credit	Actual finance-side credit entry.
Validation rule	Total PayRx/PG/DRx/HRx value less PG charges and taxes should equal total bank credit/net settlement received.

11.2 Final Doctor / Lab Payout

> *"Final payout = Net received amount + Benefit applied + Commercial / PG charges applied."*

Component	Meaning
Total appointment booking amount	Gross payable value for the appointment or service.
Benefit applied	Voucher, Healthcoin, claim, or other benefit value.
Commercial / PG charges	Commercial or payment gateway charge component.
Net received amount	Actual amount collected after benefits and charges.
Final payout	Net received plus benefits plus commercial/PG charge components.
Payout rail	RazorpayX for doctors and labs/providers.

12. Product Solution Architecture

Layer	Capability
Data ingestion layer	API ingestion from all sources, with manual upload retained as fallback.
Raw data layer	Visible stored data from PayU, PayRx, RazorpayX, and related systems.
Proofing layer	Unique order ID and amount validation for every row.
Matching layer	Transaction count matching, transaction amount matching, benefit composition matching, PG settlement matching, and payout matching.
Exception layer	Matched, partial match, proposed match, open, and manual review queues.
Manual review layer	Action paths for failed/reverted payments, processing payments, invalid VPA/account/UPI, beneficiary bank downtime, and manual processing.
Governance layer	Role-based edits, audit logs, before/after values, update reason, maker-checker approval.
Payout layer	RazorpayX payout triggering for doctor and lab payouts.
Dashboard layer	Counts, amounts, percentages, current open items, and open item trends.

13. Dashboard and Reporting Capabilities

The dashboard was designed as an operational cockpit for reconciliation and settlement monitoring, not simply a static report.

Dashboard Dimension	Details
Payment sources	PayU, PayRx, RazorPay, RazorpayX, and related payout visibility.
Status buckets	Matched, Partial Match, Proposed Match, Open, Total.
Measures	Count and amount.
Percentages	Matched %, Partial Match %, Proposed Match %, Open %.
Open item trend	Last 7 days, 30 days, 3 months, 1 year.
Raw data visibility	Stored ingested data from PayU, PayRx, RazorpayX, and other sources.
Ingestion methods	API ingestion and manual upload through UI.
Export	Data export from screens retained for operational use.

13.1 Raw Data Proofing and Issue Visualization

- Every row should have a unique order ID.
- Every entry should have an amount associated with it.
- Problematic rows should be highlighted.
- Issue count should be clearly visible at the top of the screen.

- Clicking a highlighted row should show import date, import time, import method, and API response for authorized users.

14. Manual Review, Audit, and Controllership

A strong part of the HealthPay product scope was controlled manual review. The system did not assume all exceptions would automatically resolve. It created a governed path for manual investigation, correction, approval, and auditability.

Manual Review Area	Required Handling
Transaction count mismatch	Show extra transactions, show source, move to manual review.
Transaction amount mismatch	Apply exception criteria; if unresolved, highlight and move to manual review.
Failed / revert payment	Review audit log and process manually if required.
Processing payment	Alert on failure or completion.
Invalid VPA/account/UPI	Trigger communication to concerned RM through PayRx or manually.
Beneficiary bank down	Retry.
Edited raw data	Show count of edited records and list editor, last updated date/time, update reason, and detailed audit log.

14.1 Governance Controls

- Edit permissions were role-based.
- Every manual correction was audit-logged.
- Maker-checker approval existed for edited raw data and manual payout.
- Audit logs included update date, update time, updated by, value before update, value after update, and comments.
- API response visibility was restricted to authorized users.

15. SOP and Operating Model

The project also had a defined operating rhythm. Samadhan participated in daily operations review until stabilization, which ensured that the product worked in live financial operations and not only in product documentation.

SOP Area	Final Interpretation
Product and operations ownership	Samadhan led the full project as Lead PM and participated in daily operations review until stabilization.
Daily review	Recon and payout issues were reviewed operationally until the automated process stabilized.
Manual upload fallback	Retained to support resilience when API ingestion or source availability became constrained.
Failure fallback	Manual processes and stakeholder communication paths existed for trusted and untrusted PayRx/system failure scenarios.
Settlement continuity	Operating model supported payout and settlement continuity even when exceptions occurred.

15.1 Risk Management

Risk	Mitigation Plan
Resource absence	Next level to pick up the task.
System failure where PayRx can be trusted	Execute end-to-end recon manually.
System failure where PayRx cannot be trusted	Communicate to stakeholders, bypass PayRx using PayU, reconcile PayU inflow to RazorpayX outflow, and settle transactions manually.

16. Delivered Scope

Capability	Final Status
Recon and settlement dashboard	Went live

Capability	Final Status
Raw data ingestion	Went live
API ingestion	Went live for all sources
Manual upload	Retained as fallback
Matching engine	Went live
Manual review workflow	Went live
Audit log	Went live
Payout triggering	Went live
Doctor payouts	Went live
Lab payouts	Went live
RazorpayX payout rail	Went live as final payout rail
Role-based edits	Went live
Maker-checker controls	Went live for edited raw data/manual payout

17. Success Metrics and Business Impact

17.1 Primary Success Metrics

Metric	Why It Mattered
Payout TAT	Measured how quickly doctors and labs/providers received payouts after reconciliation.
Settlement accuracy	Measured correctness of reconciliation between payment, benefit, settlement, bank credit, and payout records.

17.2 Secondary Metrics

Metric	Why It Mattered
Open item count and amount	Measured unresolved financial exposure.
Manual review volume	Measured exception load and automation gap.
Payout failure count	Measured provider payment issue volume.
Settlement variance	Measured PG/bank/finance mismatch risk.
Raw data edit count	Measured source-data correction frequency and control risk.
Provider escalations	Measured provider-facing payout friction.

17.3 Confirmed Impact

- Manual reconciliation effort was eliminated because the process became completely automated.
- Payouts happened on T+1.
- Payout delays and provider escalations reduced.
- Settlement accuracy improved.
- Audit, compliance, and controllership visibility improved.
- No before/after numeric metrics are available, so exact percentage improvement should not be claimed.

> *"Use the phrase "enabled T+1 payouts and eliminated manual reconciliation effort" rather than inventing a percentage impact."*

18. Your Role and Ownership

Samadhan was the Lead Product Manager for the full HealthPay Recon & Settlement Automation project.

Ownership Area	Details
Product leadership	Lead PM for the full project.
Product artifacts	Owned PRD, user stories, and dashboard design.
Solutioning	Owned end-to-end solutioning across pay-in reconciliation, payout automation, exception workflows, audit controls, and payout triggering.
Operations involvement	Participated in daily operations review until stabilization.
Cross-functional alignment	Worked across HealthPay, PayRx, HRx, DRx, Labs, Finance, payment gateways, and payout operations.

Ownership Area	Details
Governance design	Defined role-based edit permissions, audit logs, and maker-checker controls.

19. Interview Narrative

In 2023, immediately after Service Guarantee, I led the HealthPay Recon & Settlement Automation project as Lead Product Manager. The goal was to automate pay-in and pay-out reconciliation across PayU, RazorPay, PayRx, HRx, DRx, Labs, Voucherify, Healthcoin, Claims, Finance, and RazorpayX.

The problem was that HealthPay transactions were not simple cash payments. A single transaction could include customer payment, voucher, Healthcoin, claim benefit, PG charges, settlement entries, and final doctor or lab payout. These components existed across different systems, making manual reconciliation effort-heavy, slow, and risky.

I owned the PRD, user stories, dashboard design, matching logic, manual review flows, audit controls, and payout-triggering requirements. I also participated in daily operations review until stabilization.

The final solution included API ingestion from all sources, manual upload fallback, raw data visibility, matching engine, exception handling, manual review, audit logs, role-based edit permissions, maker-checker approval, dashboarding, and RazorpayX payout triggering for doctors and labs.

The reconciliation process became completely automated, payouts moved to T+1, and the project improved settlement accuracy, payout governance, auditability, compliance, and financial controllership.

20. Portfolio and Resume Version

20.1 Portfolio Summary

> *"Led HealthPay Recon & Settlement Automation in 2023 as Lead PM, delivering API-led reconciliation, matching engine, manual review, audit logs, maker-checker controls, dashboarding, and RazorpayX payout triggering across pay-in/pay-out flows for doctor and lab payouts. The initiative eliminated manual reconciliation effort, enabled T+1 payouts, reduced provider escalations, and improved settlement accuracy and financial controllership."*

20.2 Resume Bullet

> *"Led HealthPay Recon & Settlement Automation as Lead PM, delivering API-led reconciliation, matching engine, manual review, audit logs, maker-checker controls, dashboarding, and RazorpayX payout triggering across pay-in/pay-out flows for doctor and lab payouts; eliminated manual reconciliation effort and enabled T+1 payouts with stronger settlement accuracy and controllership."*

20.3 One-Line Framing

> *"HealthPay transformed reconciliation from manual cross-system checking into a controlled financial operations layer for payment matching, benefit validation, settlement proofing, payout triggering, exception management, and audit-ready reconciliation."*

21. Appendix: Key Rules and Data Elements

21.1 Key Validation Rules

Validation Area	Rule
PayU vs PayRx	Match successful/captured transaction count and transaction value using PayRx transaction ID / PG Order ID.
RazorPay vs PayRx - full PG payment	Match count and amount using PG Order ID.
RazorPay vs PayRx - partial PG payment	RazorPay order amount should equal PayRx order amount minus Voucherify, HealthCoin, and Claim.
Benefit-only payment	DRx benefit details should match Voucherify, Healthcoin, and

Validation Area	Rule
	Claims using unique parameters.
PayRx vs DRx/HRx	Match appointment booking value, benefit value, and net value using PayRx transaction ID.
PG settlement vs Finance	Net settlement amount after PG charges and taxes should match bank credit.
Doctor/lab payout	Final payout equals net received amount plus benefit applied plus commercial/PG charges.

21.2 Important Raw Data Fields

Source	Representative Fields
PayRx	Order Date, Order Time, Order ID, PayRx Order ID, Transaction Type, Settlement ID, PG Order ID, UTR, Customer Name, Customer Mobile, Provider ID, Provider Name, Doctor Name, Doctor Fees, Order Amount, Claims Discount, Voucher Applied Amount, Voucher Code, Healthcoin Amount, Net Paid By Customer, Payment Method, Payout Status, Transfer Amount, Reason for Failure.
RazorpayX	Payout ID, account number, amount, currency, mode, purpose, status, UTR, contact ID, fund account ID, failure reason, notes, fees, tax, contact name, reversal ID, created/processed/reversed/scheduled timestamps, fee type.
Voucherify / Healthcoin	Customer name, phone number, voucher/benefit code, status, amount, redeem date, SRC_ID.